

1.
A: E, H
B: A
C: F, G
D: A, E
E: C
F: D, E
G: B, E
H: D

[A, E, 1]
[A, H, 1]
[B, A, 1]
[C, F, 1]
[C, G, 1]
[D, A, 1]
[D, E, 1]
[E, C, 1]
[F, D, 1]
[F, E, 1]
[G, B, 1]
[G, E, 1]
[H, D, 1]

2.
DFS called for vertex A
Vertex A visited and received the stamp 0
Current array: [0, -1, -1, -1, -1, -1, -1, -1]

DFS called for vertex E
Vertex E visited and received the stamp 1
Current array: [0, -1, -1, -1, 1, -1, -1, -1]

DFS called for vertex C
Vertex C visited and received the stamp 2
Current array: [0, -1, 2, -1, 1, -1, -1, -1]

DFS called for vertex F
Vertex F visited and received the stamp 3
Current array: [0, -1, 2, -1, 1, 3, -1, -1]

DFS called for vertex D
Vertex D visited and received the stamp 4
Current array: [0, -1, 2, 4, 1, 3, -1, -1]
E already visited
Go back to C

DFS called for vertex G
Vertex G visited and received the stamp 5
Current array: [0, -1, 2, 4, 1, 3, 5, -1]
E already visited

DFS called for vertex B
Vertex B visited and received the stamp 6
Current array: [0, 6, 2, 4, 1, 3, 5, -1]
A already visited

DFS called for vertex H
Vertex H visited and received the stamp 7
Current & Final array: [0, 6, 2, 4, 1, 3, 5, 7]